# Yihao Xu

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## Research Profile

- Computational design science researcher working on multimodal prediction for crowdfunding and multi-view clustering for industrial fault diagnosis.
- First author of a paper at the *International Conference on Information Systems* (2025) and an article in *IEEE Sensors Journal*; the ICIS paper was a Best Complete Paper Nominee / Best-in-Track Paper.
- Experience in neural model design, joint optimization, temporal adaptation, and LLM-based assessment of organizational culture; Python, PyTorch, and SQL.
- Teaching assistant across two fall terms; coordinated six study sessions involving 60 participants who are deaf or hard of hearing in an AR accessibility study.

## Education

**M.S. in Management Science and Engineering** **Expected 2027**
*School of Management, University of Science and Technology of China (USTC)* *Hefei, China*
Advisor: Prof. Hefu Liu.

**B.M. in Information Management and Information Systems** **2024**
*School of Management, Hefei University of Technology (HFUT)* *Hefei, China*
Selected coursework (HFUT grades):
Linear Algebra (96/100, A), Operational Research (95/100, A), Business Data Analytics (91/100, A).

## Research Interests

**Topics:** Digital platforms and crowdfunding · Cultural and creative markets · Organizational culture · Accessible human–AI interaction
**Methods:** Multimodal representation learning · Multi-view deep clustering · Temporal generalization · Computational text measurement

## Published Research

**1. Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction**
**Yihao Xu**, Wenxiang Zhao, and Xiaoyu Yao.
*46th International Conference on Information Systems (ICIS)*, Sharing Economy, Platforms, and Crowds Track, Nashville, TN, 2025. *ICIS 2025 Best Complete Paper Nominee / Best-in-Track Paper.*

- Creative projects convey cultural meaning through text, images, and metadata. This study models those values and examines their roles in crowdfunding success.
- The Cultural Value-based Prediction Model (CV-PM) maps content features to Throsby's six cultural values. Convolutional networks reduce noise in each value representation, and attention combines the values for prediction.
- On 14,539 Kickstarter projects, AUC reached 94.72% in Art and 95.14% in Design, exceeding the strongest respective baselines by 1.49 and 3.65 percentage points.
- In value ablations, the largest accuracy losses occurred when removing social value in Art (7.17 percentage points) and aesthetic value in Design (5.19).

**2. Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint**
**Yihao Xu**, Mingfeng Lu, Gang Wang, and Fang Fang.
*IEEE Sensors Journal*, 25(11), 19921–19936, 2025. doi:10.1109/JSEN.2025.3560460.

- In unsupervised fault diagnosis, features learned separately from clustering may not suit the clustering task. Multiple sensors also provide consistent and complementary information that the model needs to retain.
- MsDCN uses a sensor-aware attentive encoder to learn consensus representation features and separate encoders to retain sensor-specific features. A consensus-specific constraint guides joint optimization of reconstruction and clustering.
- Across 20 runs per dataset, MsDCN achieved the highest mean clustering accuracy on six gearbox and bearing datasets. Gains over the strongest comparison method were 5.74–11.35 percentage points on the two DIRG bearing datasets. Ablations evaluated the sensor-aware attentive encoder and consensus-specific constraint.

## Conference Presentation

**Yihao Xu**, Wenxiang Zhao, and Xiaoyu Yao. *Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction.* 46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, Dec. 2025.

## Work in Progress

**3. Multimodal Learning under Evolving Environments for Crowdfunding Outcome Prediction**

**Yihao Xu**, Wenxiang Zhao, and Hefu Liu. *Planned submission to Information Systems Research in 2026.*

- Campaign text and images contain both overlapping and complementary information. SSC-TAP retains shared, modality-specific, and interaction information within a common semantic reference and learns their joint contribution to prediction before launch.
- Content distributions and the predictive relevance of that content can change over time. Temporal relevance adaptation models these changes separately and adjusts each semantic dimension's predictive contribution for the target campaign and its launch time.

**4. From Anchor to Evidence: A Theory-Driven Model for Newly Launched Crowdfunding Success Prediction**

Wenxiang Zhao, **Yihao Xu**, and Xiaoyu Yao. *Working Paper.*

- Uses anchoring-and-adjustment theory to distinguish summary cues on listing pages from information on project detail pages when predicting crowdfunding success before launch.

**5. When Differences Drive Synergy: Cultural Complementarity as a Resolution to the Efficiency–Resilience Paradox in Supply Chains**

Along Liu, Mengru Zhang, **Yihao Xu**, and Hefu Liu. *Research in progress.*

- Investigates cultural complementarity in the trade-off between supply chain efficiency and resilience. Uses an LLM-based method to assess cultural matching between firms from corporate disclosures.
- The measurement approach draws on organizational culture scales and examples selected by experts. Comparisons with suppliers and customers account for disclosed transaction shares and support analyses over time and during disruption and recovery.

**6. Knowing When to Be Firm: Communication Style Shifts in AI-Assisted Food Delivery**

*Early-stage Research Project.*

- Food-delivery customers may request coupons while leaving emotional needs unspoken. The project examines shifts between assertive and nonassertive AI language during coupon assistance and emotional support. It considers customer preference profiles and responses across conversational turns in relation to subsequent purchase decisions.

## Research and Teaching Experience

**Research Assistant**, University of Science and Technology of China *Jun. 2026–Present*

Supported an AR smart-glasses study on communication access and social inclusion for people who are deaf or hard of hearing; coordinated materials, participant arrangements, device setup, and six study sessions involving 60 participants.

**Teaching Assistant**, University of Science and Technology of China *Fall 2025, 2026*

Information Technology and Organizational Strategy 42 students; 40 hours

Instructor: Prof. Juan Qin; coordinated assignments, group activities, and feedback.

## Awards and Scholarships

**Awards**

- Outstanding Graduate of Anhui Province (top 3% students) 2024
- Outstanding Student Award (top 10% students) 2023

**Scholarships**

- China National Scholarship 2025
- First-Class Graduate Scholarship, USTC 2025, 2026
- Shishengqing Scholarship, School of Management, HFUT 2024
- Academic Progress Scholarship, School of Management, HFUT 2022
- First-Class Scholarship, HFUT (top 3% students) 2021

## Internships

**Research Intern** *Jun.–Aug. 2026*

*USTC Yangtze River Delta Science and Technology Strategic Frontier Research Center*

Conducted desk research, case analysis, and comparative studies of firms' early development and commercialization in commercial spaceflight, AI, and semiconductors.

**Technical Service Intern, CEO Office**, Volkswagen (Anhui) Automotive Co., Ltd. *Apr.–Jul. 2024*

*Hefei, China*. Built and maintained an automotive after-sales database for recurring service reporting and operational analysis.

## Computational Skills and Languages

**Computational Methods:** Multimodal learning · Multi-view deep clustering · Attention mechanisms · Temporal generalization · Sentence embeddings and semantic retrieval

**Programming:** Python · PyTorch · SQL · Stata

**Languages:** Chinese (native) · English (IELTS Academic 7.5)